Siemens Ethernet protocol driver

Overview

Adroit supports the Siemens Ethernet protocol in the following 2 ways:

1. ISO over TCP/IP for the S7-200/300/400 PLC

Using the following PLC cards, tested so far :

S7-200 CPU 226 with CP Card CP 243 – 1

See Appendix B at the end of this document to see how to setup an Ethernet connection on the S7-200 PLC's using MicroWin version 4.0 that allows ADROIT to communicate to it.

S7-315 2DP using the following cp cards : (S7-317 2DP probably the same , not confirmed yet).

CP343-1-Lean Siemens partnumber 6GK7 343-1CX00-0XE0

CP343-1- Siemens partnumber 6GK7 343-1EX20-0XE0

(S7-317 2DP)Without cp cards, a direct connection into the cpu ethernet port make sure Dest TSAP CPU and RACK possitions are both set to 2.

S7-414-2DP using the following cards :

CP443-1- Siemens partnumber 6GK7 443-1EXE0-0XE0

CP443-1- Siemens partnumber 6GK7 443-1EX02-0XE0

The system requires no other drivers , stacks or plc code to run , provided the cp card is capable of communicating ISO over TCP/IP and is configured with an appropriate ethernet address and mask and in the same range as the adroit machine.

The driver can also communicate to the PLC using the IBH Link S7/S7++ connection module. See Appendix C on how to setup the communication using the IBH Link!

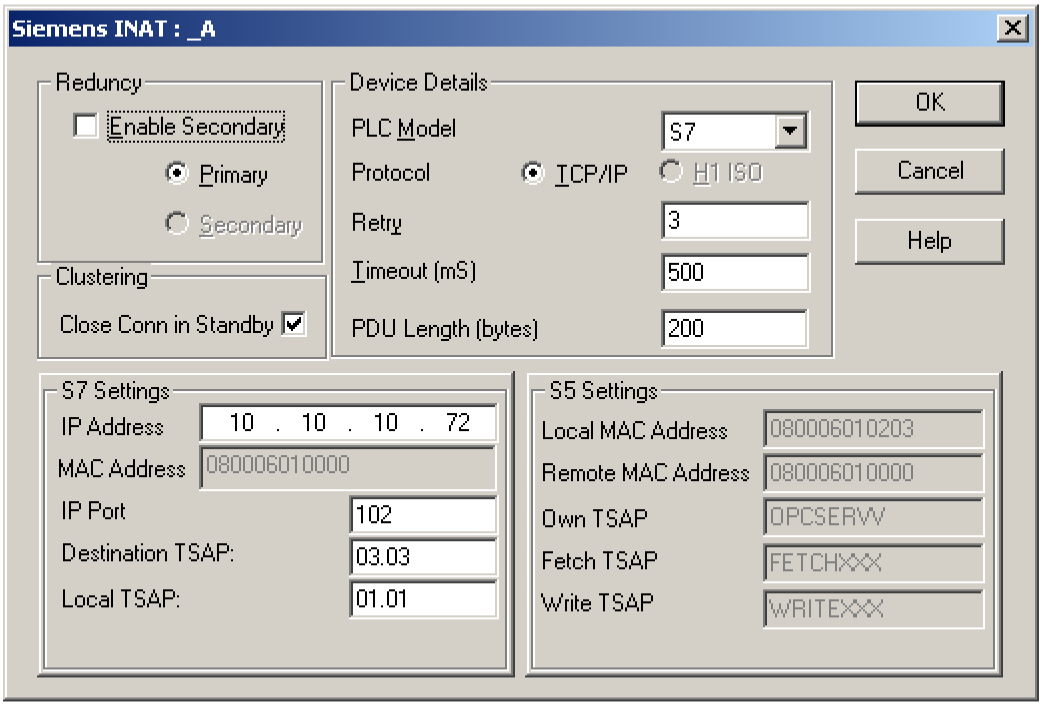
2. H1 for the S5 and S7 PLC

Using the following PLC cards, tested so far :

CP1430.

The system requires additional setup to be done on the S5 side as well as PLC code, before setting up this driver. provided the cp card is capable of communicating H1 and is configured with an appropriate Mac address in the same range as the adroit machine. See Appendix A for a step by step procedure on how to get H1 running for S5 using this driver.

Device Configuration



Redundancy

Enable Secondary – (Unchecked by default) Check if redundancy is required.

Primary - Radio selected , displays primary configuration.

Secondary - Radio selected , displays secondary configuration.

Clustering

Close Conn in Standby – (Unchecked by default) In the case of having a Clustered Environment, the Device will disconnect from the PLC when switching from Master to Standby which allows the new Master Server Device to take the same connection! This will only work in a shadow scanning cluster!

Device Details

PLC Model – S5 or S7 Selectable - (S7 default)

Protocol – Select either TCP/IP or H1, depending on protocol being used. TCP/IPP is not available when selecting S5 PLC model.

Retry - The number of retries in the case of a communication time-out or interruption.

Timeout - The time in milliseconds to wait for a transaction to the PLC to complete successfully.

PDU Length - The maximum bytes to read in a single transaction.

S7 Settings

This block will only be active when S7 PLC Model is selected.

Ethernet Address - Configurable Ethernet Address. Will only be active if TCP/IP protocol is selected.

MAC Address - Configurable MAC Address. Will only be active if H1 protocol is selected.

IP Port – Usualy 102 , change not recommended.

Destination TSAP – The Destination(PLC) TSAP works in the following manner. The TSAP format is 01.02.(Default is 03.03)

The first byte(01) is the Service Number or also known as Connection Resource number. The number represents the following:

01 – STEP 7 Connection

02 – SCADA Connection to S7-200

03 – SCADA Connection to S7-300/400

The second byte(02) represents the CPU destination where the 0 is the rack number and the 2 is the slot number.

Default values per PLC type will then most likely look like this:

S7-200 – 02.00 – As S7-200 does not have RACK and CPU position it is thus 00.

S7-300 – 03.02 – The PSU takes slot 1 and the CPU takes slot 2.

S7-400 – 03.03 – The PSU takes slot 1 and 2 and the CPU slot 3.

See Appendix C on how to setup the communication using the IBH Link to determine the TSAP settings when using IBH Link module!

Local TSAP – Deafult is 01.01. It does not have much meaning but it is the standard. It works fine with S7-300 and S7-400 but does not work for S7-200 as MicroWin requires you to setup an Ethernet connection where you have to specify the PC TSAP and it does not allow for 01.01 to be entered. For S7-200 02.01 works fine as MicroWin Allows that setting in the connection setup. See Appendix B at the end of this document to see how to setup an Ethernet connection on the S7-200 PLC's using MicroWin version 4.0.

See Appendix C on how to setup the communication using the IBH Link to determine the TSAP settings when using IBH Link module!

S5 Settings

This block will only be active when S5 PLC Model is selected.

Local MAC Address - Configurable MAC Address for the PC. It can be any spare MAC address as long as it is in the same range as the PLC network.

Remote MAC Address - Configurable MAC Address for the PLC. This is the MAC address of the PLC we will connect too.

OwnTSAP – An 8 Character String as setup in PLC. See Appendix A, Step 4. (Default is OPCSERVV).

FetchTSAP – An 8 Character String as setup in PLC. See Appendix A, Step 4. (Default is FETCHXXX).

WriteTSAP – An 8 Character String as setup in PLC. See Appendix A, Step 4. (Default is WRITEXXX).

Wiring

Please see standard Ethernet wiring documentation.

Supported addresses

Siemens S5 and S7 PLC data address types supported

For Data Blocks:

Device	Min Range	Max Range	PLC Data Type	BOOL	INT	REAL	STR
Data Block	DB0.DBX 0.0	DB 65534.DBX 65534.7	Digital	x			
Data Block	DB0.DBB 0	DB 65534.DBB 65534	Unsigned 8		X	X	
Data Block	DB0.DBW 0	DB 65534.DBW 65534	Unsigned 16		X	X	
Data Block	DB0.DBI 0	DB 65534.DBI 65534	Signed 16		X	X	
Data Block	DB0.DBL 0	DB 65534.DBL 65534	Signed 32		X	X	
Data Block	DB0.DBD 0	DB 65534.DBD 65534	Unsigned 32		X	X	
Data Block	DB0.DBF 0	DB 65534.DBF 65534	IEEE 32			X	
Data Block	DB0.DBS 0 S2	DB 65534.DBS 65534 S79	Character String				X
Data Block	DB0.DBB 0 S2	DB 65534.DBB 65534 S79	Pascal String				X

For Markers:

Device	Min Range	Max Range	PLC Data Type	BOOL	INT	REAL	STR
Memory	MX 0.0	MX 65534.7	Digital	x			
Memory	MB 0	MB 65534	Unsigned 8		X	X	
Memory	MW 0	MW 65534	Unsigned 16		X	X	
Memory	MI 0	MI 65534	Signed 16		X	X	
Memory	ML 0	ML 65534	Signed 32		X	X	
Memory	MD 0	MD 65534	Unsigned 32		X	X	
Memory	MF 0	MF 65534	IEEE 32			X	
Memory	MS 0 S2	MS 65534 S79	Character String				X
Memory	MB 0 S2	MB 65534 S79	Pascal String				X

For Inputs/Outputs:

Device	Min Range	Max Range	PLC Data Type	BOOL	INT	REAL	STR
Memory	I/QX 0.0	I/QX 65534.7	Digital	x			
Memory	I/QB 0	I/QB 65534	Unsigned 8		X	X	
Memory	I/QW 0	I/QW 65534	Unsigned 16		X	X	
Memory	I/QI 0	I/QI 65534	Signed 16		X	X	
Memory	I/QL 0	I/QL 65534	Signed 32		X	X	
Memory	I/QD 0	I/QD 65534	Unsigned 32		X	X	
Memory	I/QF 0	I/QF 65534	IEEE 32			X	
Memory	I/QS 0 S2	I/QS 65534 S79	Character String				X
Memory	I/QB 0 S2	I/QB 65534 S79	Pascal String				X

- A Character string reads character for character and return it as a string. Pascal String is the standard Siemens string which has 2 bytes in front of the actual string data bytes which contains the character ‘ P ‘ as first byte and the string length in the second byte.

Message protocol

Driver datascope displays include a listing of the parameters passes to the INAT layer, and only the RAW DB data sent/returned in each transaction.

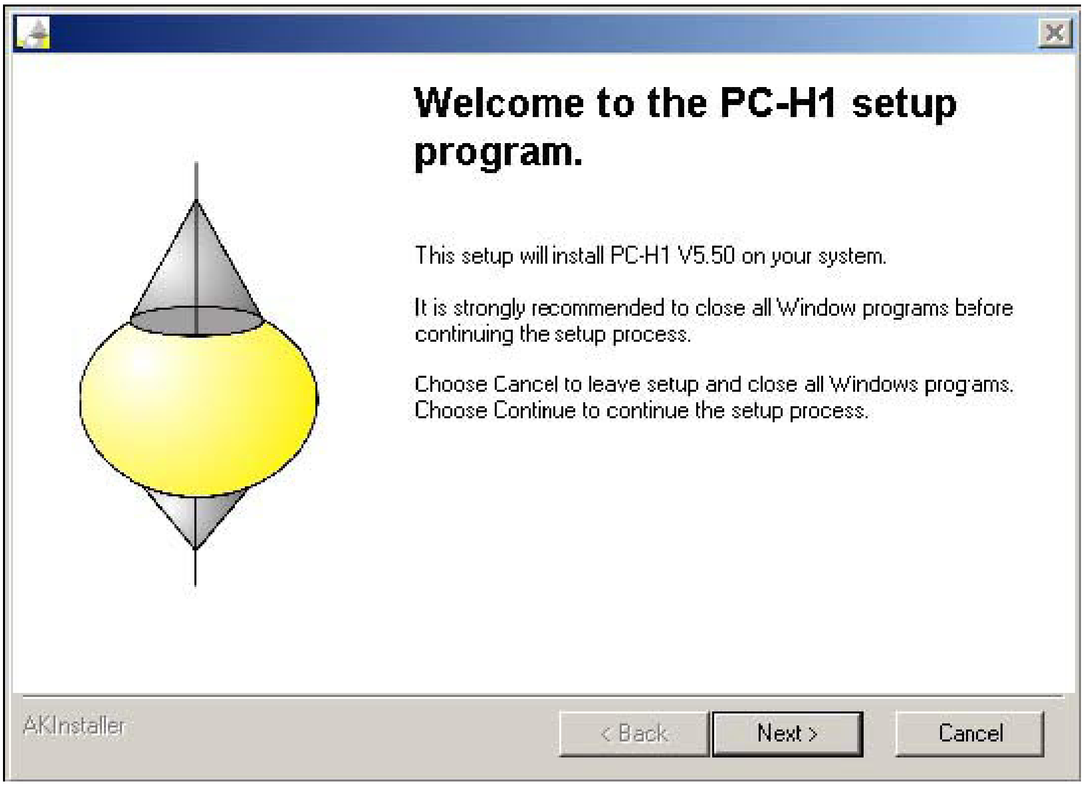
Appendix A – Installing H1 Network on the PC

Step 1 - Install H1 Driver:

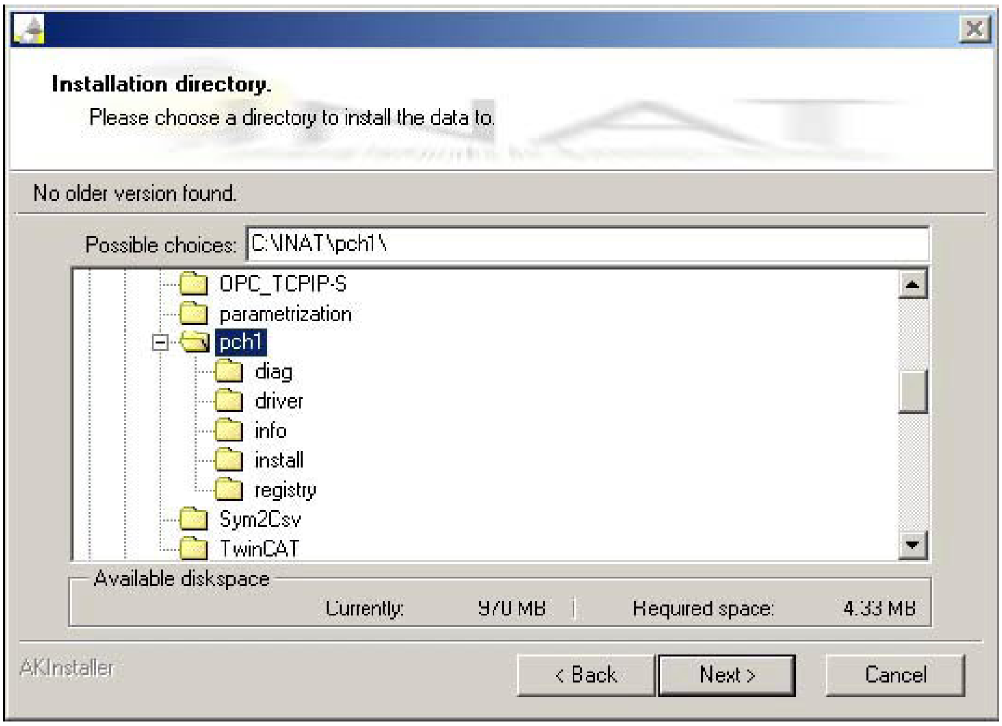
● Start pch1.exe.

● Select the Setup language and press 'OK'.

● The start dialogue window for PC-H1 setup appears.



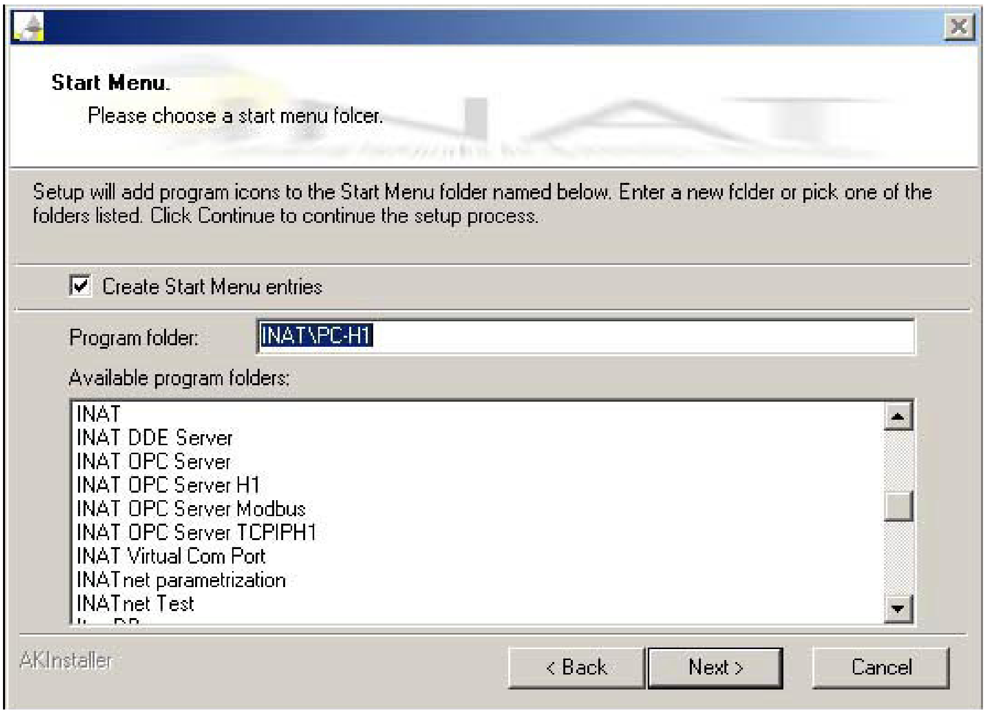
- Select 'Next'.
- The license agreement appears.
- Agree to the conditions set forth by clicking 'Yes'.
- Select 'Next'.



● Select the directory, where the files shall be stored. Default **C:\Programs\INAT\pch1\.**

● Select 'Next'

● Select the start menu folder.

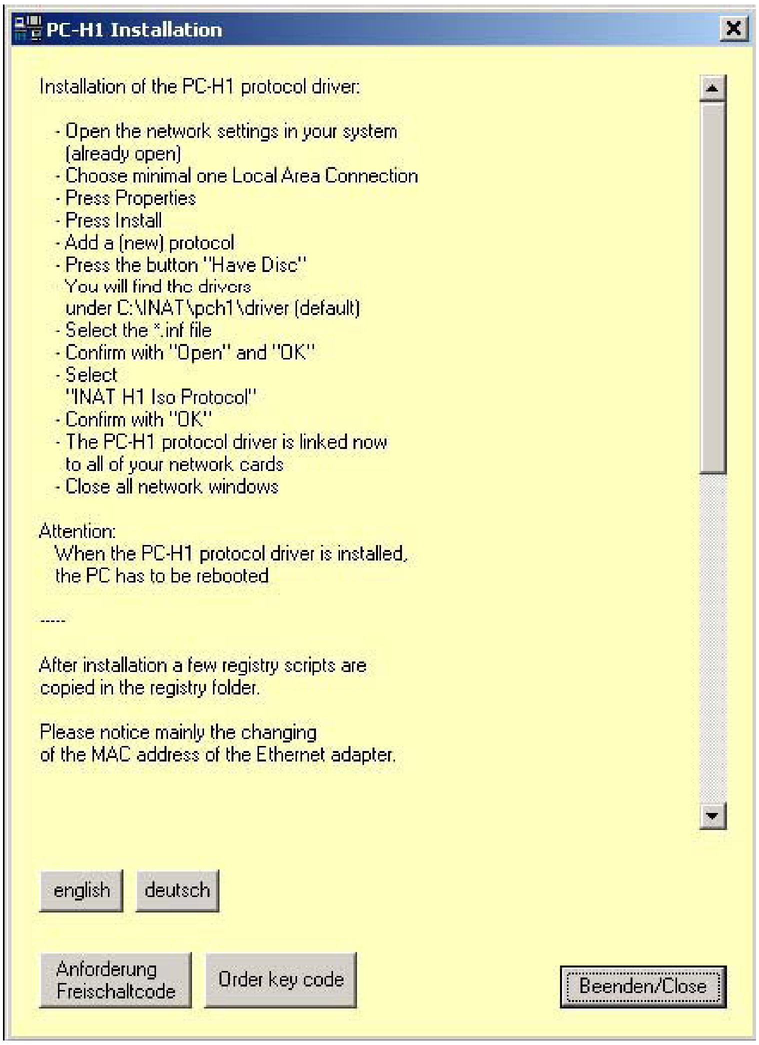


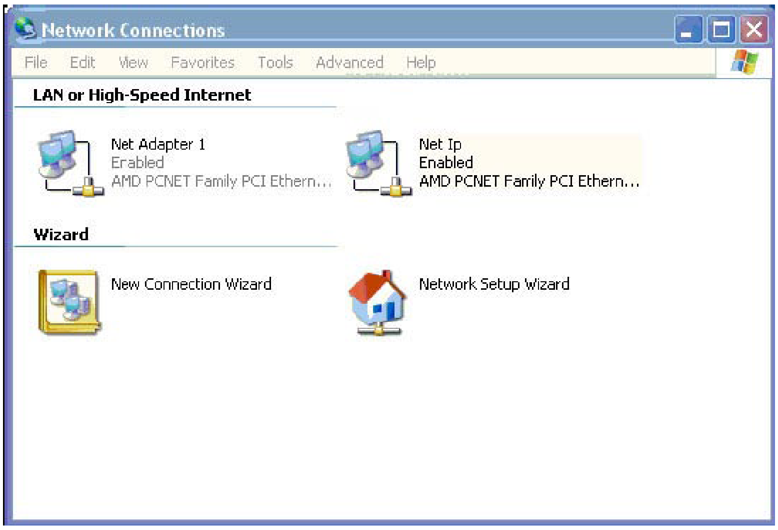
● Select 'Next'

● Installation is complete after a few seconds.

Step 2 - Install the H1 Protocol on the Network Card:

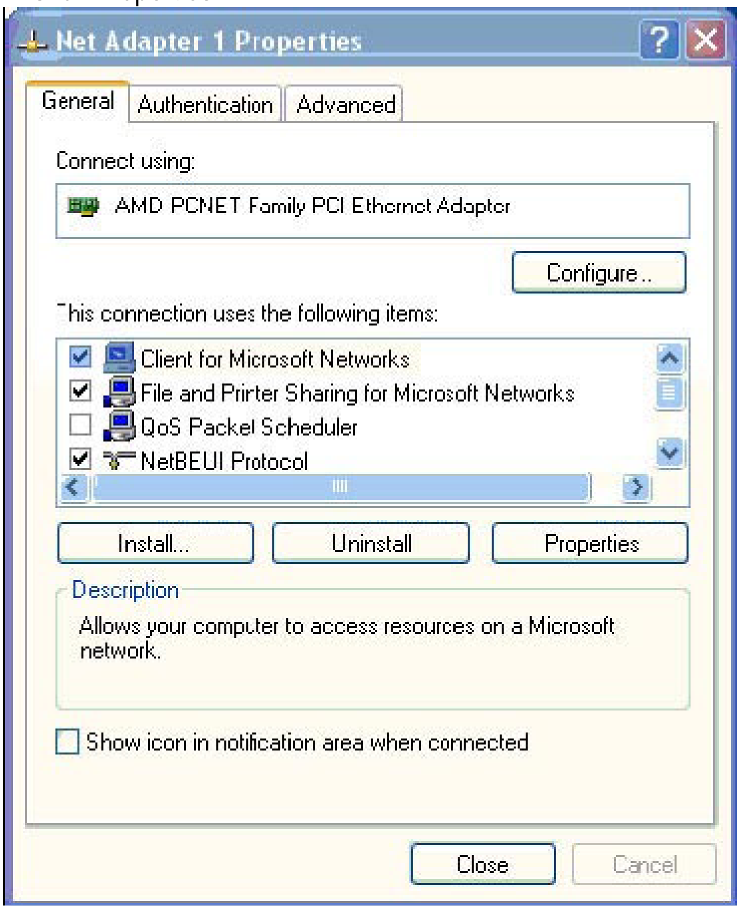
● After installation of PC-H1, the 'network settings' window opens together with the description below:



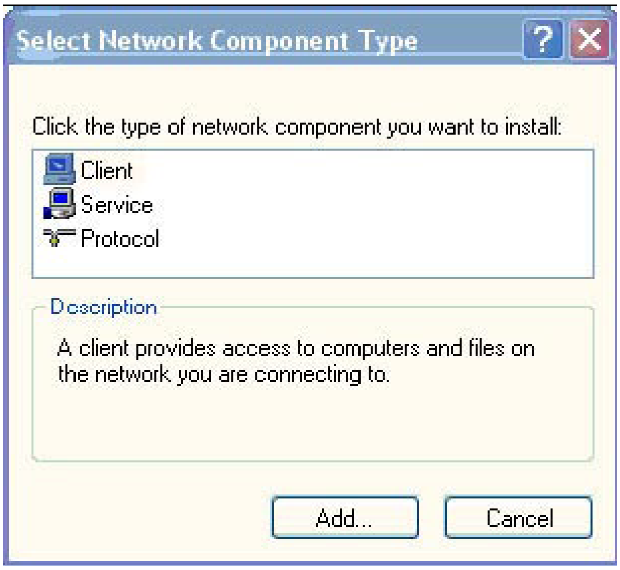


● Click Close and Double-click the network adapter that you want to configure. The following window appears.

● Click Properties.

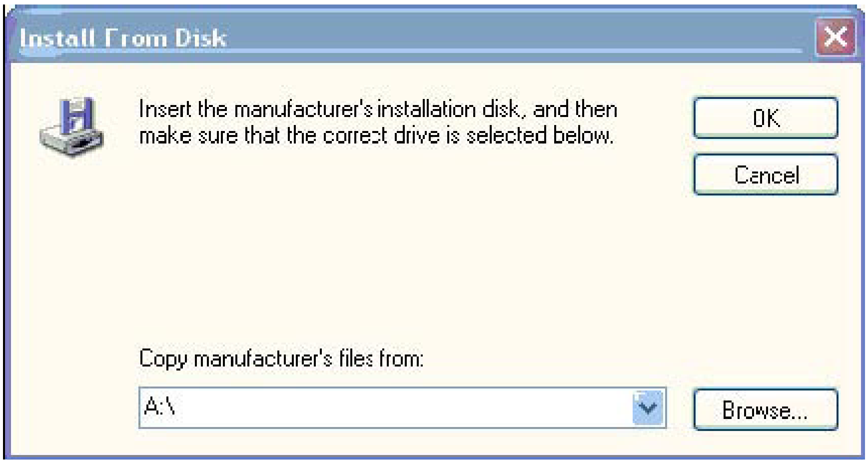


● Click 'Install'.



● Select Protocol and click Add.

● Click 'Have Disk...'

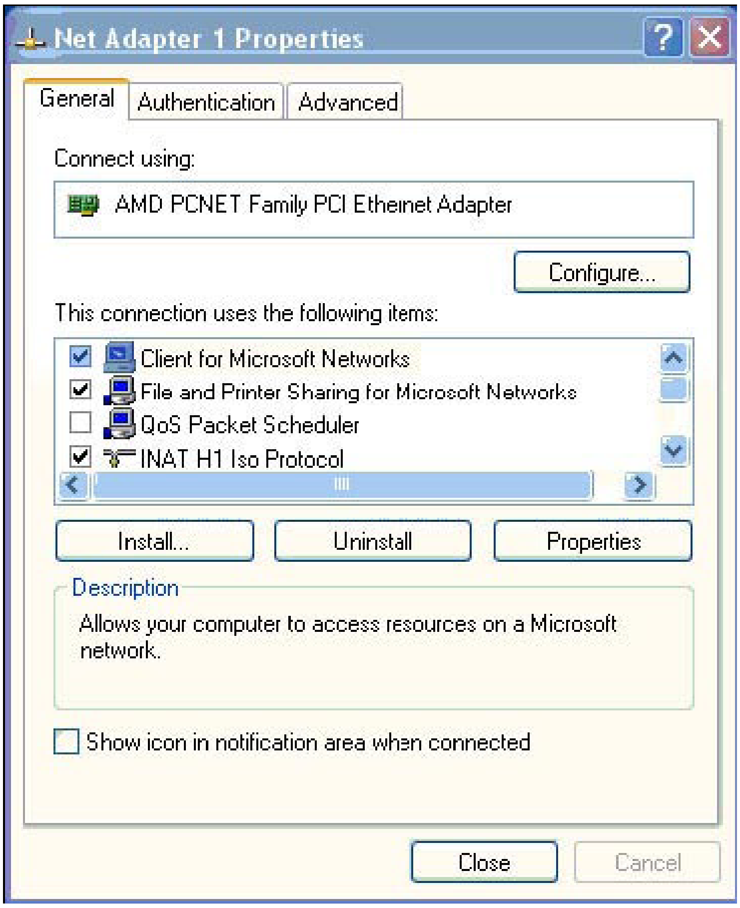


● Browse to the path of the PC-H1 installation directory, which typically is c:\Programs\INAT\pch1\driver).

● Click Open.

● Select 'INAT H1 ISO Protocol' and click OK.

● In the window 'Net Adapter properties' the H1 protocol is indicated now as below.



The QoS Packet Scheduler has to be deactivated!
In the window 'Net Adapter properties' there must not be a tick in front of the element QoS Packet Scheduler as shown above.

● Run the registry key 'start/programs/inat/PC-H1/H1 registry scripts/H1ChangeAddress.reg'.

● Start your computer again, and make sure that all programs are loaded without error messages when the system boots.

Step 3 - License the H1 protocol:

The H1 Protocol needs to be licensed as it only has a demo period of 10 hours to run. To License it do the following:

● Get the MAC Address of the required Network adapter by going to run and typing 'cmd'. In the command prompt window type 'ipconfig /All'. Write Down the MAC Address and e-mail it to 'drivers@adroit.co.za'.

● You will receive a key code as soon as possible. Open a file called 'H1free.dat' situated in 'C:\winnt\system32\drivers'. Replace the code received with the dummy code inside and save and close.

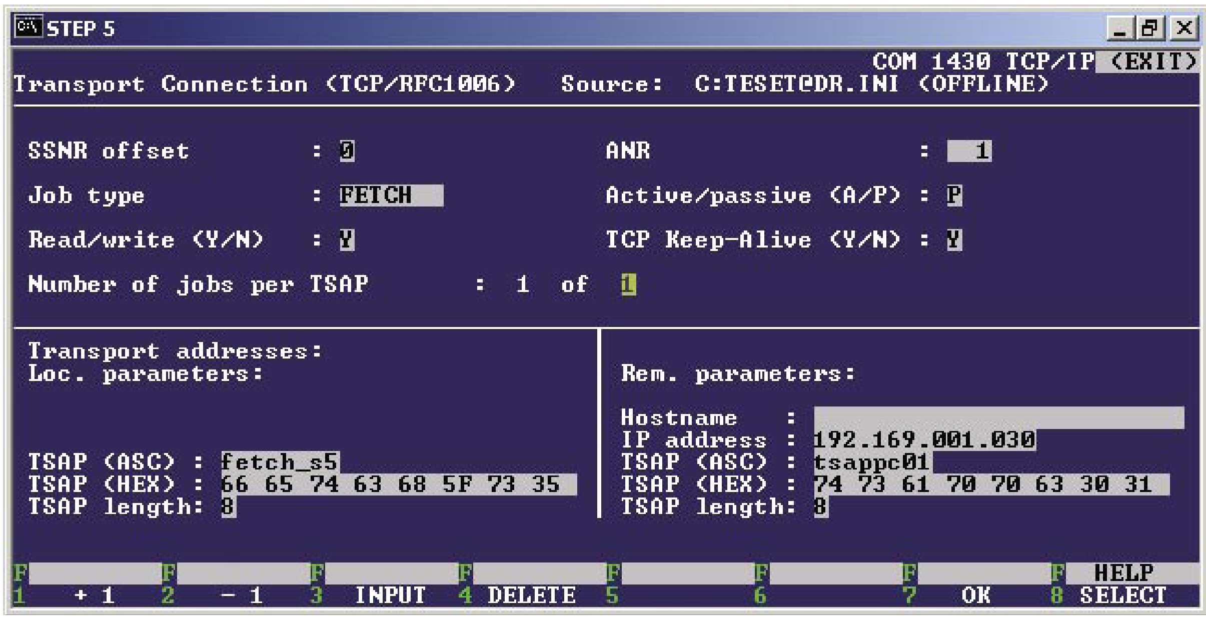
● Start your computer again, and make sure that all programs are loaded without error messages when the system boots.

Step 4 - Configuring PLC to be H1 ready:

● You need to call the SYNC function block at startup and the SEND/REC ALL function blocks in the cyclic part of your program. These are:

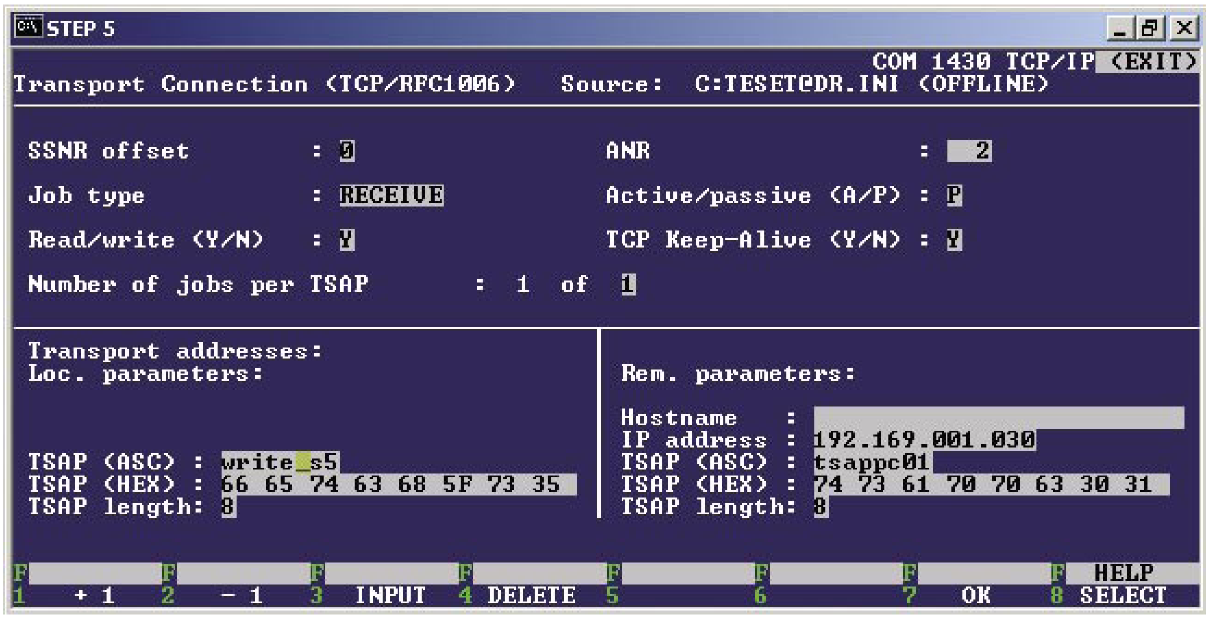
PLC	135/150/155	115
SYNC	FB125	FB249
SEND ALL	FB126	FB244
REC ALL	FB127	FB245

In Step 5 for S5 PLC – H1 communication you have to setup 2 Connections. The first is a FETCH connection and should look like the image below. The Local Parameter TSAP is unique to the FECTH connection and by default FETCHXXX. We keep the length to a standard 8 characters, but it does not have to be 8 characters long. The Remote Parameter TSAP is the TSAP of the PC or in other words the Driver. This can be any name, also standard to 8 characters and by default we make it OPCSERVV.



The second is a RECEIVE connection and should look like the image below. The Local Parameter TSAP is unique to the RECEIVE connection and by default WRITEXXX. We keep the length to a standard 8 characters, but it does not have to be 8 characters long. The Remote Parameter TSAP is the TSAP of the PC or in other words the Driver. This can be any name, also standard to 8 characters and by default we make it OPCSERVV.

The Remote TSAP MUST BE THE SAME for the FETCH and RECEIVE connections, although the Local TSAP's must be unique. The IP Address in the remote parameter box should be MAC address in our case and it indicates the MAC address of the PC and should be the same for both connections.



After creating these 2 connections your setup on the PLC side is completed and so also the setup of H1. All you need to do now is setup a new device in adroit setup.

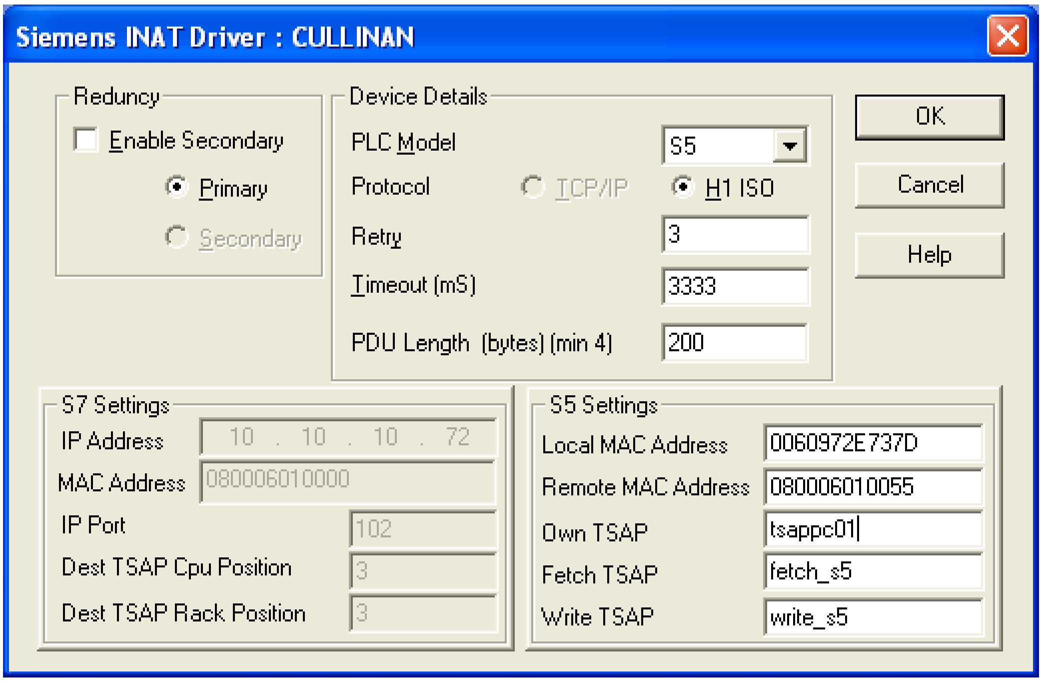
Using the setup for the S5 PLC, as it is done with the images above as an example, we setup adroit device as it is in the image above. The S5 PLC connections is setup as follow:

FETCH TSAP – fetch_s5

RECEIVE TSAP – write_s5

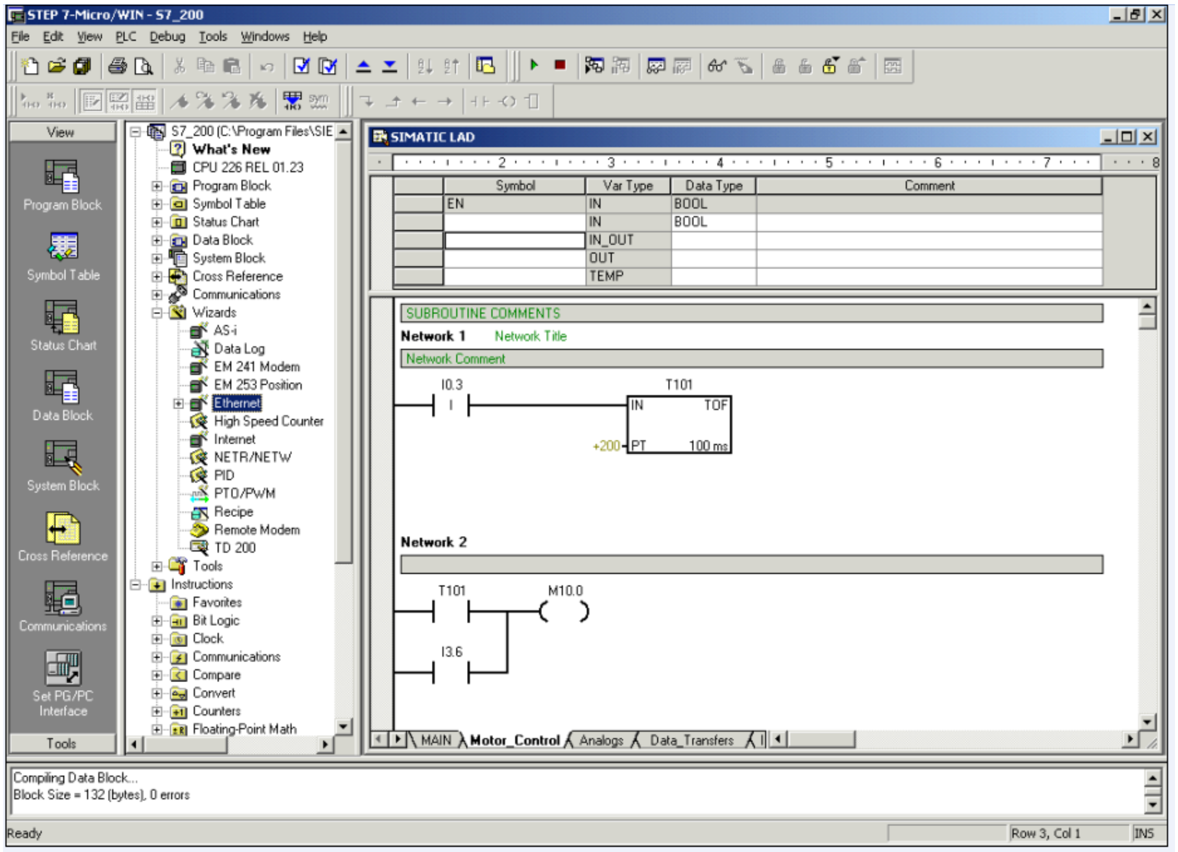
In both connections the remote TSAP is tsappc01 and the MAC address is 00 60 97 2E 73 7D.

Using this information to complete the S5 Settings in the ADROIT device driver the settings will be as in the image below where remote MAC address = PLC MAC addr.

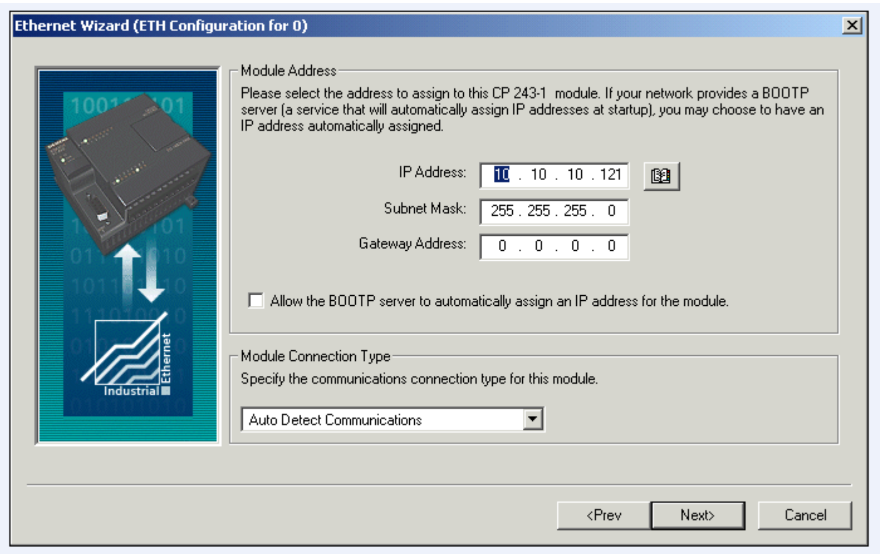


Appendix B – Creating S7-200 Ethernet Connection

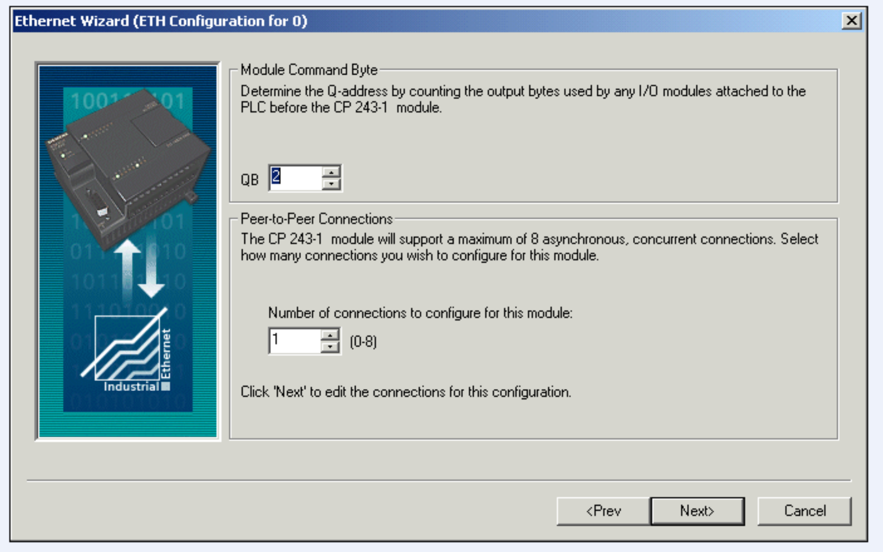
Create a new Ethernet Configuration



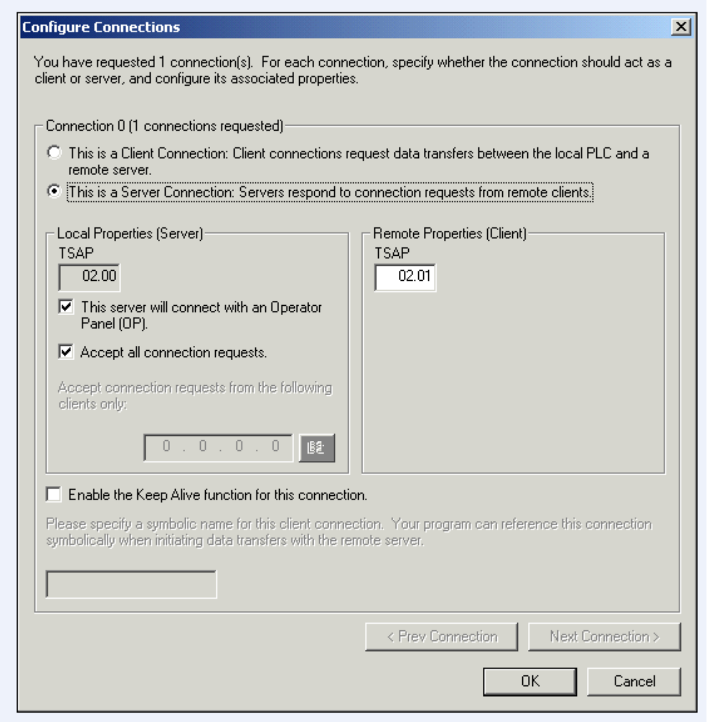
When using MicroWin 4, right click with the mouse on the Ethernet Wizard as shown above. Select 'New Configuration'. The next page will popup.



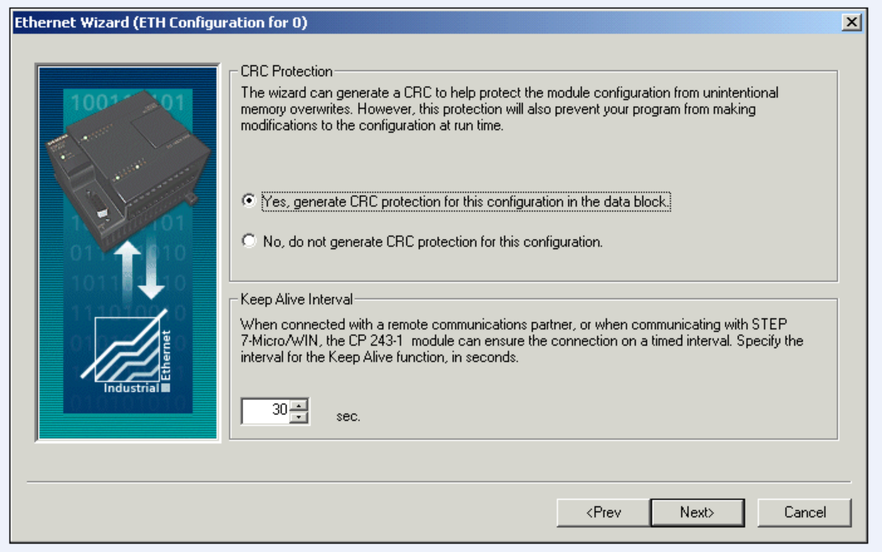
Specify the Ethernet module IP Address and then press next. The next page will popup.



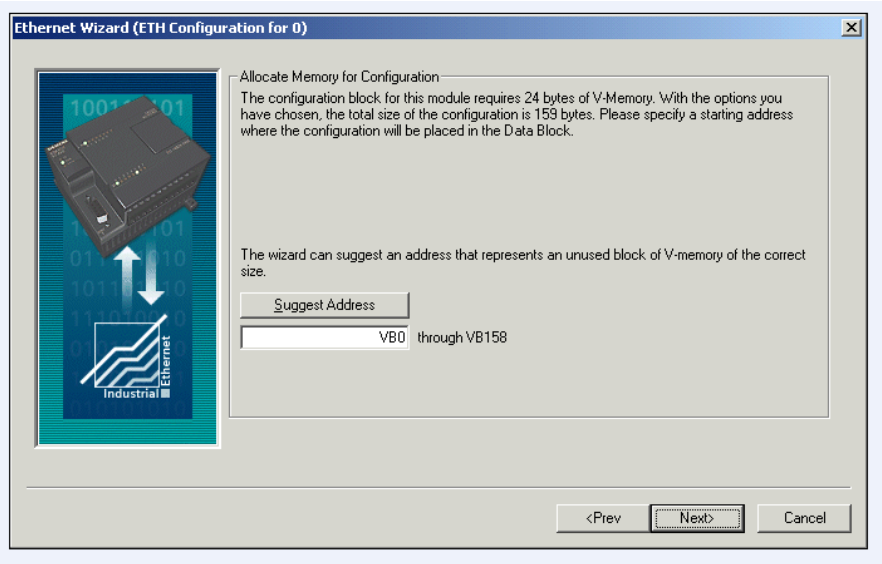
Specify the number of connections required. At least 1 for ADROIT connection. Now press next button to open the next screen.



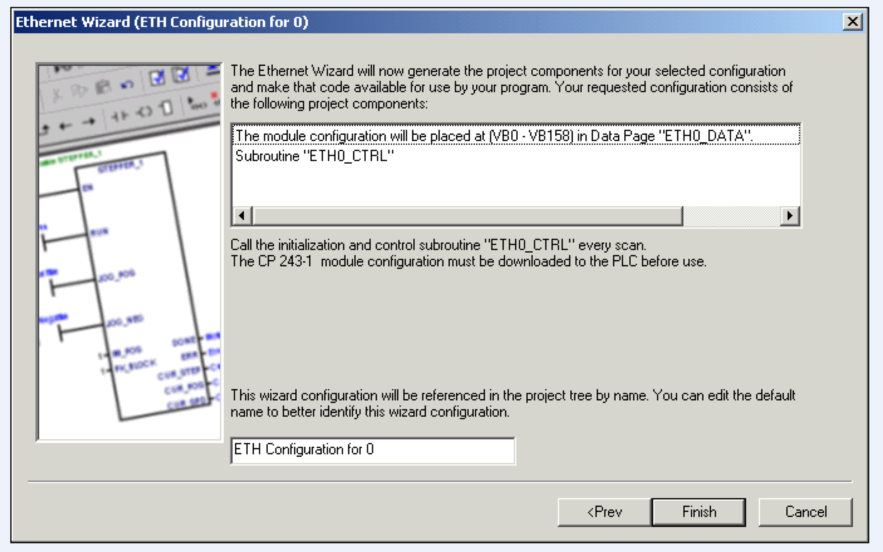
This is the most important screen of all. This is where we setup the connection too allow ADROIT to communicate to the PLC. The screen above is the exact setting required to make the communications work. We want a server connection that allows the PLC to operate as the server and ADROIT as the client which connects and requests data from the PLC. The Local TSAP on this page refers to the PLC TSAP settings. This setting is not changeable and when 'The server will connect with an Operator Pane(OP)' is selected the TSAP will change to 02.00, so the Destination TSAP setting in the driver setup must be 02.00. Also make sure 'Accept all connection requests' is checked. This will allow for any IP address to create a connection to the PLC. The Remote TSAP properties refer to the PC TSAP settings or as we know it in the driver setup as local TSAP. We prefer the TSAP of 01.01 as the local TSAP but the MicroWin does not allow this, so it was changed to 02.01 which means your local TSAP setting in the driver setup must be 02.01. Once ready click the OK button to continue to the next screen.



Generate CRC protection, it is a safer option. Click next button to continue to the next screen.



This will just allocate memory area for the configuration, leave as default. Click next button to continue to the next screen.



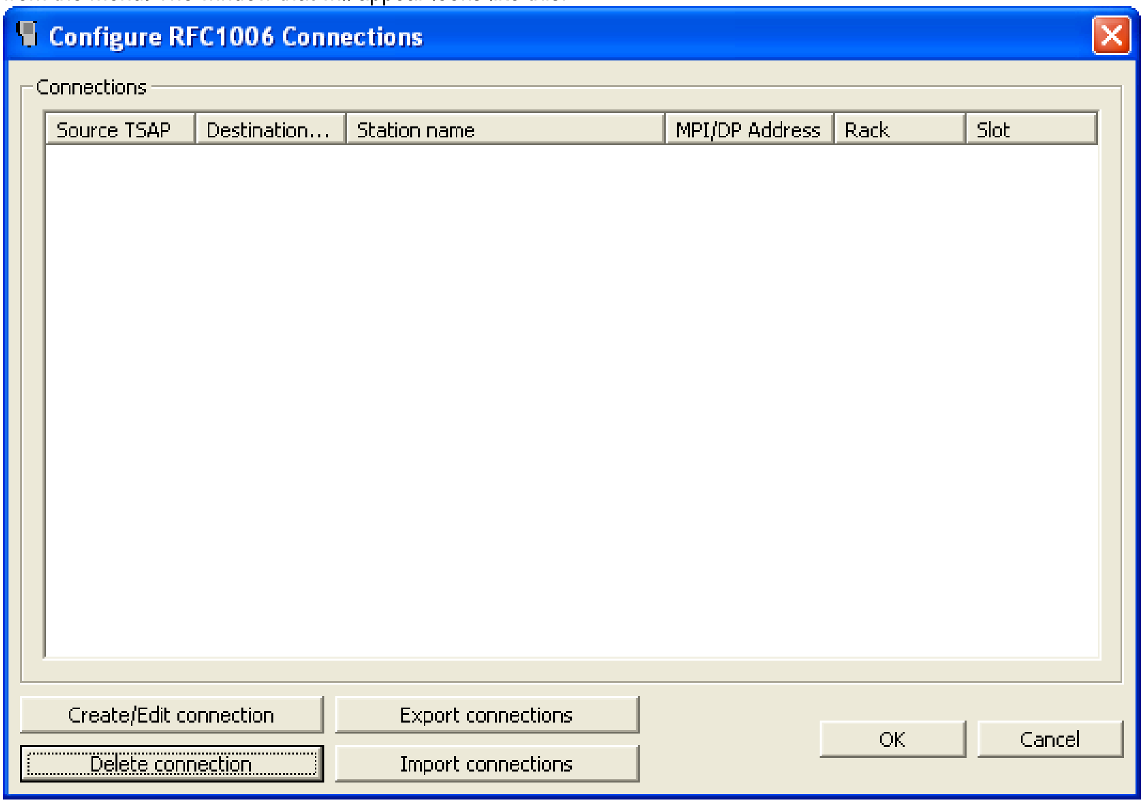
This is the last screen. Here you can give a name for your configuration if so do required, but leave it as default otherwise. Press finish button to complete the new Ethernet connection setup. After this setup is completed make sure to press the 'save all' button. Now go to the PLC menu and press 'Compile All' item. This will compile and check everything. As long as there is no errors, press download button to download to the PLC. ADROIT should be able to communicate to the PLC!

Appendix C – Creating connection to IBH Link S7/S7++ module

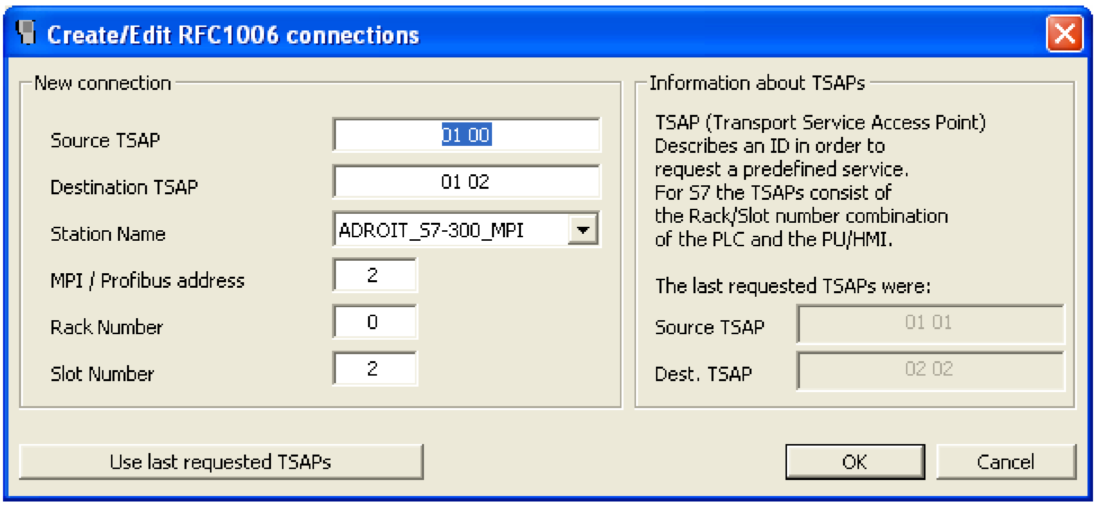
For communication to occur between the driver and an IBH Link S7/S7++ unit, the following steps will be needed!

1. On the same server machine or on another server machine in the network, the application called “RFC1006 to IBHLink converter” need to be started. This application is supplied with the software you receive when you purchased the IBH link module. “RFC1006 to IBHLink converter” application listens on port 102 and it might be possible that one of siemens software already listens on port 102 and the application will fail to start. You can end the process called “SIMATIC IEPG Help Service(s7oiehsx.exe)”! This will allow the “RFC1006 to IBHLink converter” application to start up.

2. Now you need to configure a connection inside the “RFC1006 to IBHLink converter” application’s configuration window, which can be found by clicking on the tray icon representing the application and selecting ‘Configuration’ from the menu. The window that will appear looks like this:



Click on ‘Create/Edit connection’ button to add a new connection. You should have a window looking like this.



The Source TSAP is the same as what is specified on the drivers Local TSAP, by default the driver uses 01.01, so we can make the settings the same.

The Destination TSAP is the TSAP that will be used to relay to the PLC from this application. This TSAP is calculated in the following manner. The connection resource is 02 and the RACK/CPU slot position is the actual MPI/PROFIBUS address of the PLC. In this case the MPI address of the PLC I am trying to connect to is 2, hence we have a TSAP of 02.02! Use the same TSAP in the drivers Destination TSAP setting.

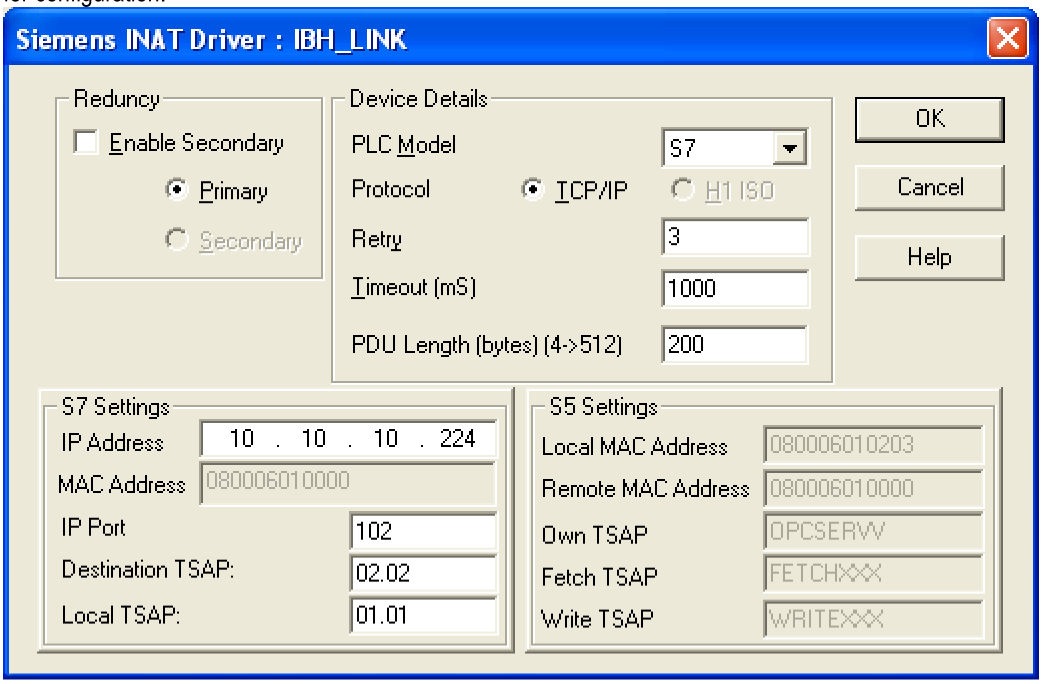
The Station Name is a preconfigured IBHNet connection setting that was created initially to enable access to the unit and being able to program the PLC via the IBH Link unit. If this is not done yet please do it first using the "IBHNet and IBHLink settings" utility also supplied with the unit. Select the correct station that will be used to connect to the PLC!

Lastly specify the MPI/Profibus address of the unit you want to access and also the CPU's RACK and SLOT number.

You should have the same settings as the image above, except for the station name, if the MPI address is 2.

Press OK and OK again. Configuration of new connection is now completed.

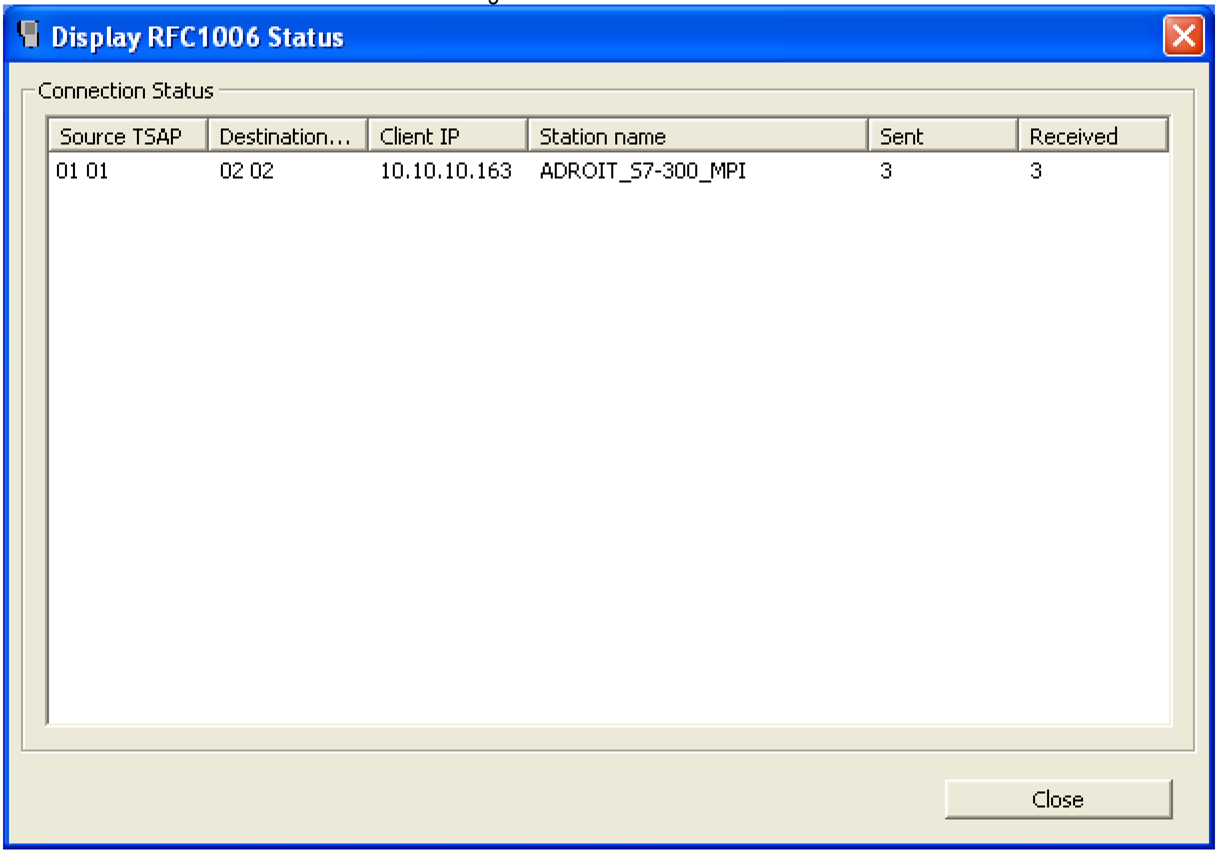
3. On the driver dialog specify the TSAP's as explained in step 2 and specify the IP Address of the server machine that the "RFC1006 to IBHLink converter" application is running on. This can be local or remote. See image below for configuration.



4. The driver should be able to communicate to the PLC using the IBH Link module through MPI/Profibus.

5. There is a 'Status' monitor tool that can be used to see if any connection is active on the "RFC1006 to IBHLink converter" application. Click on the tray icon representing the application and select 'Status' from the menu. When

a successful connection is established the following can be seen in the monitor.



Revisions

Date	Changes
22-May-2006	Revision 1 : First release of the driver.
23-Nov-2006	Revision 5a : Negative extension bug fixed, Booleans added.
13-Apr-2007	Revision 6 : H1 Capabilities for S5 and S7.
14-Jul-2009	Revision 20 : S7-200 Capabilities Added.
15-Jan-2013	Added info on how to connect to IBH Link module.